Automatic Passing Vehicle to Cyclist Distance Estimation

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Abstract

This paper presents initial work on the development of a novel computer vision-based tool to automatically detect the distance between a cyclist and a passing vehicle from a camera on another vehicle (either in front or behind the bicycle being viewed). In a range of test videos the average error was only 88mm, making it feasible to use to assess how closely vehicles are passing bicycles.

Keywords: Computer Vision Application, YOLO, Object Detection

1 Introduction

For safety in Ireland (and many other countries) vehicles overtaking cyclists are required to leave a minimum clearance of 1m where the speed limit is 50km/h or less and a minimum clearance of 1.5m where the speed limit is over 50km/h. However, there appear to be no statistics around how closely vehicles overtake cyclists.

The research in this paper develops a simple system which can determine the clearance distance. A camera is mounted looking forwards or backwards on a bicycle with the intention of measuring the clearance distance for a cyclist either in front of or behind the cyclist with the camera.

2 State of the Art

Traditional distance estimation methods typically rely on stereo vision or LiDAR, but monocular systems are gaining popularity due to their cost-effectiveness and simplicity. [Huang et al., 2019] proposed a robust method for inter-vehicle distance estimation using monocular vision by combining vanishing point detection with object localization. Their framework demonstrated strong performance under varying road conditions and established the value of geometric cues in single-camera setups.

Expanding on this idea, [Liu et al., 2021] integrated convolutional neural networks (CNNs) with monocular inputs to estimate distances to vehicles. Their architecture combines object detection and regression tasks, highlighting the potential of deep learning to infer spatial relationships directly from image content without explicit calibration.

[Zhu and Fang, 2019] introduced a novel learning-based approach to predict object-specific distances from monocular images by encoding both object and context features. Their network was trained on datasets where ground truth distances were known, enabling it to generalize to new scenarios. This approach aligns closely with our use of known reference objects—like license plates—as scaling factors.

[Masoumian et al., 2021] further developed this direction by integrating deep learning-based object detection with monocular depth estimation models. Their work demonstrated the importance of combining traditional computer vision geometry with learned depth predictions for improved accuracy, especially in applications such as traffic monitoring.

[Kainz et al., 2015] and [Kul and Sayer, 2022] both proposed methods for inferring object size and wall dimensions, respectively, from 2D images using reference-based scaling, such as that used in this research.

3 System Design

The system takes a video as input, smooths the images to reduce noise, uses Yolo to find people, bicycles, vehicles and licence plates, and computes the distance between a person on a bicycle and a passing car (when the wheel of the car is in line with the wheel of the bicycle.

3.1 Cyclist Detection

To detect cyclists, a YOLOv11 based object detector is employed to identify bounding boxes corresponding to the class “person” and the class “bicycle”. Since YOLO models do not natively classify a “cyclist” class, cyclist detection is inferred by locating bounding boxes where the "person" class overlaps with the "bicycle" class. Among multiple detections, the largest person bounding box in the frame is prioritized, assuming the cyclist is closest to the camera. This helps distinguish actual cyclists from pedestrians or background figures and ensures that our measurement pertains to the target cyclist. See Figure 1.

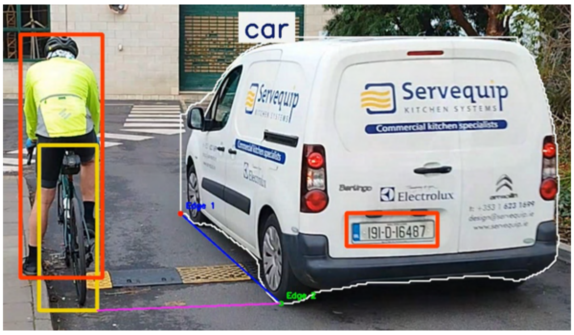

Figure 1: The bicycle (yellow box), cyclist (red box), passing vehicle (while outline), and licence plate (orange box) are all located using YOLO. The distance from the vehicle’s wheel to the bicycle/person is measured (purple line) in pixels and converted to centimetres using the size of the licence plate

3.2 Vehicle Detection

To accurately localize the vehicle, an instance segmentation model is used, which provides pixel-wise outlines instead of simple bounding boxes. This is particularly important for identifying precise points like wheel locations. A YOLOv11 variant for segmentation model is used for vehicle instance detection. The segmented output enables fine-grained localization, essential for identifying the rear or front bottom edges of the vehicle that are used for distance reference points. See Figure 1.

A pre-trained YOLOv8 license plate detection model is integrated to detect the vehicle’s license plate. To filter out false positives, a positional verification step is introduced: detected license plate bounding boxes must lie within the region of the segmented vehicle instance. This ensures that only valid plates associated with vehicles in the scene are used for scaling the pixel distances into millimetres (as licence plate width is standard). See Figure 1.

3.3 Passing Distance Estimation

Once the license plate bounding box is located, its pixel width is extracted ($Width_{pixels}$). Given that standard license plate widths for private vehicles and taxis in Ireland are approximately 520 mm ($Width_{mm}$), these numbers allow us to scale pixel distances to millimetres where the distance is roughly in the same plane. Hence we need to identify a line between the bicycle and the vehicle which meets this criteria.

We need (a) a point in line with the bottom of the bicycle which is directly below the outside of the handlebars and (b) the bottom of a wheel of the vehicle when it is in (horizontal) line with point (a). See the purple line in Figure 1. Note that the line between the points (a) and (b) is determined for every frame and the one closest to the horizontal is used for distance estimation. There is a small possibility that the box could be slightly too wide or too narrow, but this will not introduce significant error.

To locate the cyclist’s side boundary, the system identifies the end of the handlebars using the bicycle’s bounding box. The location depends on the relative position of the vehicle. If the vehicle is to the right of the cyclist, the bottom-right corner of the bicycle’s bounding box is used. If the vehicle is to the left, the bottom-left corner is chosen. This coordinate approximates the cyclist's lateral boundary for distance measurement.

The vehicle’s wheel location is determined from its instance segmentation mask. The location is based on the position of the cyclist. If the cyclist is to the left of the vehicle, the system identifies the bottom-leftmost pixel of the vehicle segmentation. If the cyclist is to the right, it extracts the bottom-rightmost pixel. This point serves as the edge of the vehicle closest to the cyclist and is used as the reference point for distance calculation.

With the reference points on both the bicycle handlebar end and vehicle wheel edge identified, the system computes the horizontal pixel distance between them ($Distance_{pixels}$). This is then converted into millimetres ($Distance_{mm}$) using the licence plate width. This estimated value represents the real-world lateral distance between the cyclist and the passing vehicle in the given video frame. See Equation (1).

$$Distance_{mm} = {\left(Distance_{pixels} \times Width_{mm}\right)}\Big/{Width_{pixels}} \qquad (1)$$

There are a number of factors which could introduce errors into the calculation, such as whether the vehicle is orthogonal to the bicycle's camera sensor plane, and the accuracy of the detection of the licence plate / the vehicle / the bicycle.

4 Evaluation

To test the system we recorded 20 videos of a cyclist being overtaken by vehicles. In order to be able to determine the actual distance between the cyclist an the vehicles, the cyclist remained still, standing over a speed bump with fixed width sections. Sample processing is shown in Figures 2 and a plot of the ground truth vs. computed distances is shown in Figure 3 along with bar chart summarising the errors in Figure 4. The mean absolute error was 88mm with a standard deviation of 108mm.

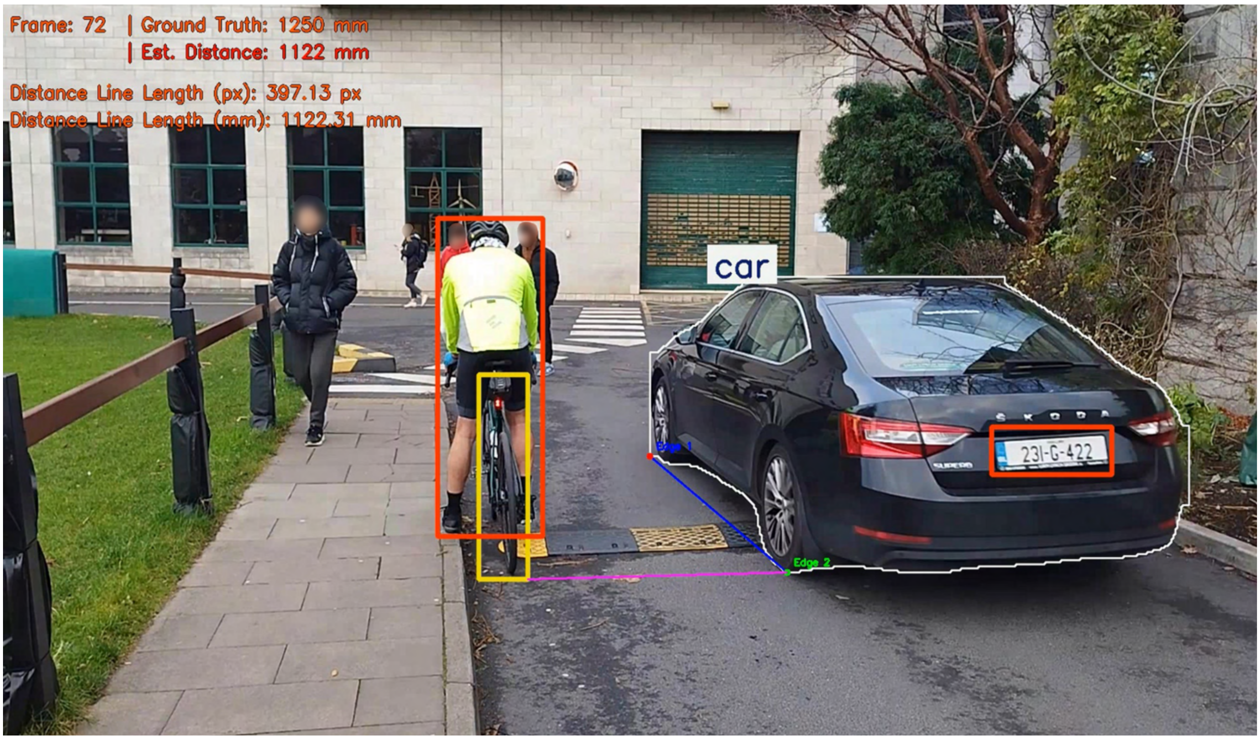

Figure 2: Sample processing and computed distances.

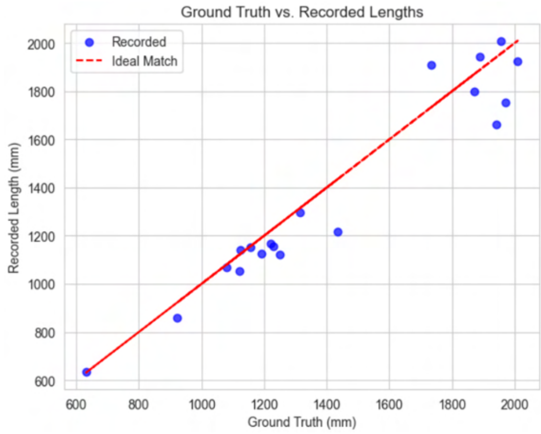

Figure 3: Distances computed between bicycles and passing vehicles (vertical) vs. the actual distance between the vehicle and the bicycle (horizontal). Each blue dot represents a single test case. The ground truth was computed by using the physical distance on the speed bumps. The diagonal red line represents perfect results so blue dots closer to the line are more accurate

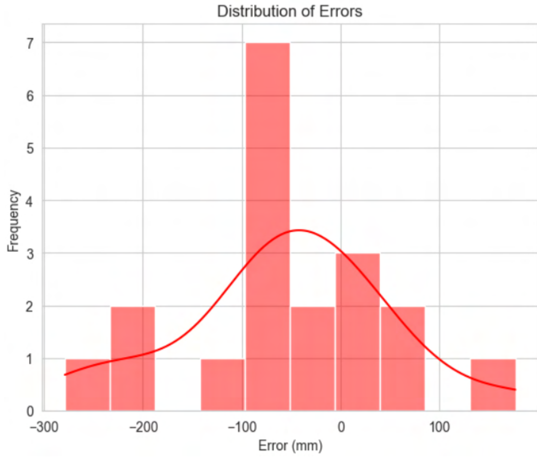

Figure 4: Bar chart summarising the errors across all test cases. The mean absolute error was 88mm with a standard deviation of 108mm

5 Conclusions

This initial research has confirmed the viability of the concept of measuring passing distance from a single uncalibrated video feed. The accuracy of the results was quite remarkable given that off-the-shelf YOLO models have been used to find the people, bicycles, cars and licence plates. We would expect further improvements in accuracy through further image processing (to more precisely locate the boundaries of objects, such as the licence plates).

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